

Project Description

Implementing Burst-MAC in LoRa Networks

Objective

The goal of this project is to implement and simulate **Burst-MAC**, a Medium Access Control (MAC) protocol designed to handle sudden burst traffic in LoRa networks.

Unlike LoRaWAN's default ALOHA protocol (which suffers heavy collisions during bursts), Burst-MAC dynamically adapts to burst conditions using grouping and semi-distributed scheduling. Students will implement and evaluate Burst-MAC using the **NS-3 LoRaWAN module**. The project requires extending the existing LoRaWAN simulation with mechanisms for burst detection, grouping, scheduling, collision resolution, and implementing Class-B mode to the NS-3 LoRaWAN module.

The students will work in a group of 3 and demonstrate the simulation in class.

Background

- **LoRa Basics:** LoRa (Long Range) is a Low Power Wide Area Network (LPWAN) technology. It uses spreading factors (SFs) and multiple channels, allowing a gateway to decode multiple transmissions if they occur on different SFs or channels. It has 3 modes of operation- Class A, B, and C.
 - **Problem:** LoRa is designed for infrequent communications, but many applications can generate burst traffic—for example, during emergencies (forest fires, volcanic eruptions, earthquakes) or in smart cities during peak hours. In such scenarios, numerous nodes transmit simultaneously, creating a sudden surge in network activity. LoRaWAN's default ALOHA-based random access lacks collision avoidance, leading to collisions, packet loss, and wasted energy.
 - **Idea:** Burst-MAC introduces a **burst mode** to handle these scenarios efficiently, while still maintaining LoRaWAN's normal behavior during regular traffic.
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An Overview of the Approach

The Burst-MAC protocol consists of the following key components:

1. Burst Detection

- Each node continuously monitors its packet generation rate. When the rate exceeds a predefined threshold, the node transitions from normal LoRaWAN Class A operation to Class B mode, enabling additional capabilities such as receiving periodic beacons from the gateway for synchronization.

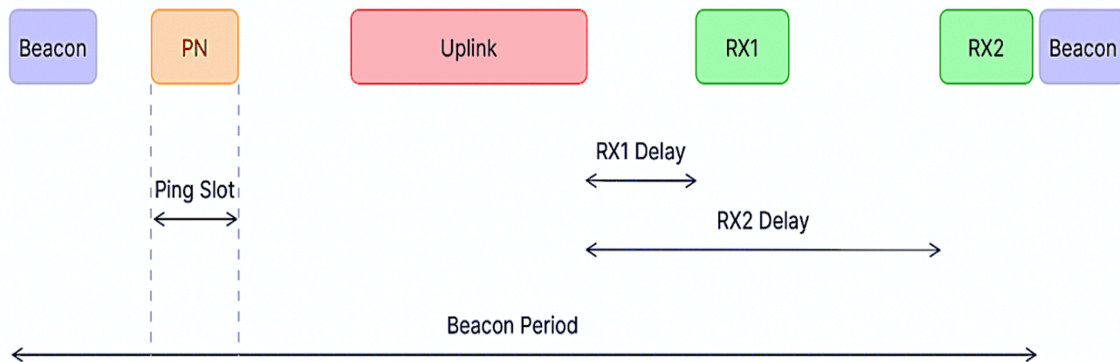


Fig: Class B mode of LoRaWAN

- Nodes notify the gateway by setting a special “burst bit” in their packets, bit is set to 1 if burst is detected, 0 otherwise.
- The gateway also detects bursts by monitoring an increase in collisions across channels.

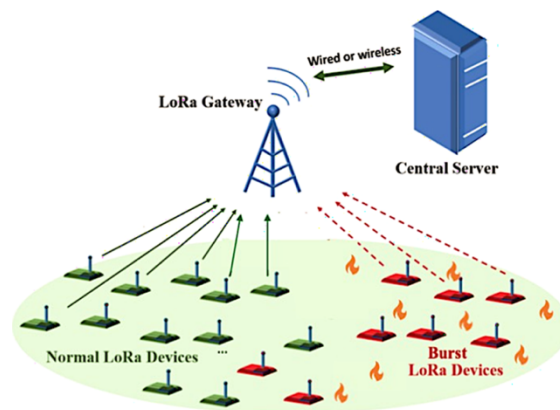


Fig: Nodes in the effected area undergo burst and switch to Burst-MAC. Nodes in non-effected areas work in LoRaWAN mode.

2. Grouping into Virtual Channels (VCs)

- Each unique (channel, SF) pair is treated as a **virtual channel (VC)**.
- Nodes are grouped by VC, meaning all nodes in a group share the same channel and SF.
- Different groups do not interfere, so one node per group can transmit concurrently.

3. Semi-Distributed TDMA Scheduling

- Within a group, nodes schedule transmissions using a **hash function** like mod function, $h(K) = k \bmod M$ (e.g., $h(\text{node_id}) = \text{node_id} \bmod \text{group_size}$).
- This maps each node to a time slot in a repeating **superframe**.
- Time slot length depends on the SF (e.g., SF7 = L, SF8 = 2L, SF9 = 3L, etc.).

4. Collision Resolution

- If two nodes hash to the same slot, the gateway detects the collision and reassigns one node to an unused slot.
- Reassignment is piggybacked onto ACK messages.

5. Downlink & ACK Handling

- ACKs are needed only during the first few rounds (to resolve collisions).
- After collisions are resolved, ACKs can be disabled to save energy, since each node already has a dedicated slot.

6. Transition Back to Normal Mode

- When the burst ends, nodes detect reduced packet rates and switch back to Class A LoRaWAN.
 - The gateway also stops broadcasting Burst-MAC schedules.
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Simulation Setup

Students can replicate the simulation in NS-3 (using the LoRaWAN module). Key setup details:

- **Network:**
 - One gateway.
 - Up to 200–1000 nodes randomly placed in a disc (radius ~6 km).
 - **Parameters:**
 - Bandwidth: 125, 250, 500 kHz
 - Coding rate: 4/5
 - Spreading factors: 7–10 (assigned based on node distance from gateway).
 - Transmission frequency in burst: one packet every 10 seconds.
 - **Burst-MAC implementation:**
 - Detect bursts using thresholds.
 - Assign nodes to VCs based on (channel, SF).
 - Apply hash-based slot allocation per group.
 - Add collision resolution logic via ACK piggybacking.
 - Implement Class-B in LoRaWAN NS-3 module.
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Detailed Description of Simulation Tasks for Students

Task 1. Baseline Setup (5%)

1. Install and configure NS-3 with the LoRaWAN module [Magrin et al., NS-3 LoRaWAN implementation].
2. Simulate a basic LoRaWAN network:
 - 1 gateway, 1000 nodes.
 - Nodes use random ALOHA-based access (default).
 - Verify simulation runs by measuring packet reception ratio (PRR).

Task 2. Burst Detection (10%)

1. Implement node-side detection: a node switches to burst mode if its packet generation rate exceeds a fixed threshold.
 - Add a “burst bit” in the uplink packet header (1 = burst, 0 = normal).
2. Implement gateway-side detection: the gateway monitors collisions on virtual channels.
 - If collision rate exceeds a threshold OR any node signals burst, gateway triggers Burst-MAC.

Task 3. Grouping into Virtual Channels (VCs) (10%)

1. Define a VC as a (channel, SF) pair.
2. Assign each node to a VC based on its current channel and SF.

3. Group nodes by VC. Nodes in the same VC interfere; nodes in different VCs do not.

Task 4. Hash-Based Scheduling (25%)

1. Define a superframe for each group = number of nodes in that group.
2. Implement hash-based slot assignment:
 - o $\text{slot} = \text{node_id} \bmod \text{group_size}$
 - o All nodes in a group compute their own slot.
3. Slot length depends on SF:
 - o SF7 $\rightarrow$ L, SF8 $\rightarrow$ 2L, SF9 $\rightarrow$ 3L, SF10 $\rightarrow$ 4L.
 - o L = base time slot duration.

Task 5. Collision Resolution (25%)

1. Gateway pre-computes schedules using the known hash function.
2. If multiple nodes map to the same slot, gateway identifies colliding nodes.
3. Unused slots in the superframe are reassigned to colliding nodes.
4. New assignments are piggybacked in ACK packets.

Task 6. Mode Switching (25%)

1. Implement gateway beaconing: gateway periodically broadcasts synchronization beacons in burst mode (Class B).
2. When node packet generation falls below threshold $\rightarrow$ switch back to Class A LoRaWAN
3. Nodes set burst bit = 0 in uplink packets to inform gateway.

Results

Metrics to Collect

- Packet Reception Ratio (PRR) = (packets received at gateway / packets sent).
- Average per-packet energy consumption (based on transmissions + retransmissions).
- Latency per packet = time from generation to successful reception at gateway.

Simulation Scenarios: Run simulations with the following variations:

1. Vary percentage of burst nodes: from 20% to 100%.
 2. Vary network size: from 200 to 1000 nodes.
 3. Vary number of VCs: e.g., 8, 16, 32, 64.
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Deliverables

1. **Code:** Modified NS-3 LoRaWAN module with Burst-MAC implementation.
 2. **Simulation scripts:** to reproduce all scenarios above.
 3. **Demonstrate in class:**
 - Code: Description of Burst-MAC components implemented.
 - Results.
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Expected Outcomes

By following this project, students will:

1. Learn how to implement MAC-level scheduling in LoRa networks.
2. Simulate and observe how grouping + hash-based scheduling reduces collisions.
3. Gain hands-on experience with NS-3 simulation of IoT networks.